Auteur: Wout Vanderborght
Studierichting: Informatica
Oefeningenreeks 6D nr. 30.23

$$\sum_{n=0}^{\infty} \frac{n!}{n^n}$$

D'Alembert:

$$\lim_{n \to +\infty} \frac{\frac{(n+1)!}{(n+1)^{n+1}}}{\frac{n!}{n^n}}$$
$$= \lim_{n \to +\infty} \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!}$$
$$= \lim_{n \to +\infty} \frac{\cancel{n!} \cdot \cancel{(n+1)} \cdot n^n}{\cancel{n!} \cdot (n+1)^{n \cancel{+1}}}$$
$$= \lim_{n \to +\infty} \frac{n^n}{(n+1)^n}$$
$$= \lim_{n \to +\infty} \left(\frac{n}{n+1}\right)^n$$
$$= \lim_{n \to +\infty} \left(\left(1 + \frac{1}{n}\right)^n\right)^{-1}$$
$$= e^{-1} = \frac{1}{e} < 1 \Rightarrow \text{convergent}$$

References